LP-III Viva Prep: DAA, ML, Blockchain

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Viva questions for SPPU BE Computer (2019 course) LP-III, unit by unit from the 410241, 410242 and 410243 syllabus, plus questions on each lab practical. Answers are short enough to say out loud.

- Group A, Design and Analysis of Algorithms: A · DAA
- Group B, Machine Learning: B · Machine Learning
- Group C, Blockchain Technology: C · Blockchain

How the viva runs

The LP-III practical exam is 50 marks: you perform one practical from any of the three groups, show the output, and then face a 5–10 minute oral. The examiner usually starts from **your practical** and moves outward into **that subject's theory**.

1. **Expect to explain your code first.** Know what each function does, why you chose that algorithm or model, and its time complexity or evaluation metric. Each group tab ends with questions on its practicals.
2. **Then come concept questions** from the unit your practical belongs to: Fibonacci → Unit II complexity, Knapsack → Unit III, N-Queens → Unit IV, KNN/SVM → ML Unit IV, smart contract → Blockchain Unit V.
3. **Answer in this order:** a one-line definition, then one example, then one advantage or limitation. Stop there and let them ask the next thing.
4. **Draw when you can.** A recursion tree, a confusion matrix, a block linked by hashes or an elbow curve on paper makes the answer obvious.
5. **If you don't know, say what you do know** that is nearby ("I haven't used Kernel PCA, but normal PCA projects data onto directions of maximum variance…"). Never bluff a definition.
6. **Keep numbers ready:** your practical's output (accuracy, R2, max profit, number of queens), the complexity of each algorithm, and the formula for each metric.

Rapid-fire: the 30 most-asked questions

If you only have an hour, learn these. Details are in each group's tab.

#	Group	Question	One-line answer
1	A	What is an algorithm?	A finite, well-defined sequence of steps that takes input and produces output in finite time.
2	A	Big-O vs Big-Ω vs Big-Θ?	Upper bound (worst-case growth) vs lower bound vs tight bound (both).
3	A	Time complexity of recursive Fibonacci?	$O(2^n)$ time, O(n) stack space; iterative is O(n) time, O(1) space.
4	A	P vs NP?	P = solvable in polynomial time; NP = a solution can be verified in polynomial time. $P \subseteq NP$.
5	A	NP-complete vs NP-hard?	NP-complete = in NP and every NP problem reduces to it (3-SAT); NP-hard = at least that hard but need not be in NP (halting problem).
6	A	Greedy vs Dynamic Programming?	Greedy makes one locally best choice and never revisits it; DP solves all overlapping subproblems and combines stored answers.
7	A	Why doesn't greedy work for 0/1 knapsack?	Items can't be split, so the best-ratio-first choice can leave capacity unused (60/10, 100/20, 120/30, W=50: greedy 160, DP 220).
8	A	Complexity of Huffman coding?	O(n log n) using a min-heap.
9	A	Backtracking vs Branch and Bound?	Backtracking is DFS that abandons infeasible branches; B&B uses bounds to prune branches that can't beat the best answer, usually for optimisation.
10	A	What is amortized analysis?	The average cost per operation over a worst-case sequence of operations (e.g. dynamic array append is O(1) amortized).

#	Group	Question	One-line answer
11	B	Supervised vs unsupervised learning?	Supervised learns from labelled data (classification, regression); unsupervised finds structure in unlabelled data (clustering).
12	B	Overfitting vs underfitting?	Overfitting = low training error, high test error (high variance); underfitting = high error on both (high bias).
13	B	Normalization vs standardization?	Min-max scales to [0,1]; standardization makes mean 0 and standard deviation 1 (z-score).
14	B	What is PCA?	Dimensionality reduction that projects data onto orthogonal directions of maximum variance (eigenvectors of the covariance matrix).
15	B	R2, RMSE, MAE?	Variance explained (1 = perfect); root of mean squared error; mean absolute error.
16	B	How does KNN work?	Finds the k nearest training points by distance and takes a majority vote (or average). Lazy, non-parametric.
17	B	What is SVM?	Finds the hyperplane with maximum margin between classes; support vectors are the closest points; kernels handle non-linear data.
18	B	Precision vs recall?	Precision = TP/(TP+FP), how many predicted positives are right; recall = TP/(TP+FN), how many actual positives were found.
19	B	Bagging vs boosting?	Bagging trains models in parallel on bootstrap samples and averages (Random Forest); boosting trains sequentially, each fixing the previous one's errors (AdaBoost).

#	Group	Question	One-line answer
20	B	How do you choose k in K-Means?	Elbow method: plot WCSS against k and pick the bend; silhouette score as a check.
21	B	What is backpropagation?	Computing the gradient of the loss for every weight with the chain rule, from output back to input, then updating weights by gradient descent.
22	B	Why do we need activation functions?	They add non-linearity; without them, a deep network collapses to one linear model.
23	C	What is a blockchain?	A distributed, append-only ledger of blocks, each holding the previous block's hash, kept consistent by consensus.
24	C	Symmetric vs asymmetric cryptography?	One shared key (AES) vs a public/private key pair (RSA, ECC).
25	C	What is a Merkle tree?	A binary tree of hashes whose root summarises all transactions in a block; proves inclusion with O(log n) hashes.
26	C	Proof of Work vs Proof of Stake?	PoW: miners spend computation solving a hash puzzle (Bitcoin); PoS: validators are chosen by staked coins (Ethereum since 2022).
27	C	Public vs private vs consortium blockchain?	Open to anyone (Bitcoin) vs one organisation controls access vs a group of organisations governs it (Hyperledger Fabric networks, R3 Corda).
28	C	What is a smart contract?	Code stored on the blockchain that executes automatically when called; deterministic and immutable once deployed.

#	Group	Question	One-line answer
29	C	What is gas?	The unit that measures computation on Ethereum; fee = gas used × gas price, paid in ETH.
30	C	What is MetaMask?	A browser-extension crypto wallet that stores your keys and lets dApps request transaction signatures.